ST121: Assignment 2

[redacted]

March 2026

1. Anderson-Darling Statistic

a)

The **null hypothesis** is the observed sample $(x_1, \ldots, x_n)$ has the hypothesized distribution with cumulative distribution function $F_X(y)$. The **alternate hypothesis** is the observed sample $(x_1, \ldots, x_n)$ does not come from the hypothesized distribution $F_X(y)$.

b)

The Anderson-Darling Statistic provides a numerical measure of how well the sample fits a proposed distribution. It is a more precise measure than the visual representation given by a QQ-Plot. In particular, it is observed that the $(F_n(y;\theta) - F_X(y;\theta))^2$ term measures the squared distance between the observed and theoretical cumulative distribution functions at every point, which quantifies the deviation between the two models. The $F_X(y;\theta)(1 - F_X(y,\theta))$ term exists to ensure the integral is weighted correctly near extreme values of the hypothetical distribution. When $F_X(y,\theta)$ is near zero or one, this term is near zero, and when $F_X(y,\theta)$ is near approximately 0.5, this term is at its maximum. Since $F_X(y;\theta)(1 - F_X(y,\theta))$ is in the denominator, it allows the tails of the distribution to be heavily weighted in the integral. Finally, the $f_X(y;\theta)$ term represents the probability density function of the hypothetical distribution, which ensures the integral is weighted appropriately. It averages the discrepancy between the observed and hypothetical distributions with respect to the hypothesized distribution across the integral.

c)

Below, there is a function that calculates the Anderson-Darling statistic from an input vector of observations.

```r
compute.ad.test<-function(xs){
  n<-length(xs)
  ordered.data<-sort(xs)
  mu<-mean(xs)
  sigma<-sd(xs)
  F_x<-pnorm(ordered.data,mean=mu, sd=sigma)
  terms=numeric(n)
  for(i in 1:n){
    term = ((2*i-1)/n)*(log(F_x[i])+log(1-F_x[n+1-i]))
    terms[i] = term}
  sum = sum(terms)
  Test_Stat = -n-sum
  return(Test_Stat)}
```

This function is applied to the provided datasets x1 and x2.

```r
load("samples_Q1.rda")
compute.ad.test(samples.Q1$x1)
```

```
## [1] 0.3774705
```

```r
compute.ad.test(samples.Q1$x2)
```

```
## [1] 1.219235
```

d)

We continue the exploration of Anderson-Darling statistics by designing an algorithm to calculate the p-value according to the given pseudo-code. The algorithm described is as follows:

```r
compute.p.values <- function(xs, B = 10000){
  n_val_1 <- length(xs)

  #calculate stats for this dataset
  mean_val_1 <- mean(xs)
  sd_val_1 <- sd(xs)

  #calc the t-obs using earlier
  t_obs_val_1 <-compute.ad.test(xs)

  #saving the t_obs values for the datasets
  T_obs_LIST_1 <- numeric(B)

  #create norm dist
  for (i in 1:B){
    #simulate sample under h0
    sim_sample_1 <-rnorm(n_val_1,mean = mean_val_1, sd = sd_val_1)

    #ad stat
    T_obs_LIST_1[i] <- compute.ad.test(sim_sample_1)
  }

  #calc p-value
  p_value_1 <-mean(T_obs_LIST_1>=t_obs_val_1)

  return(list(p_value = p_value_1, T_obs_LIST = T_obs_LIST_1, t_obs_val = t_obs_val_1))
}
```

Next, we calculate the p-value for each dataset and save it:

```r
x1_p_val_results <-compute.p.values(samples.Q1$x1)

x2_p_val_results <- compute.p.values(samples.Q1$x2)

print(paste("The p-value for x1 is", x1_p_val_results$p_value))
```

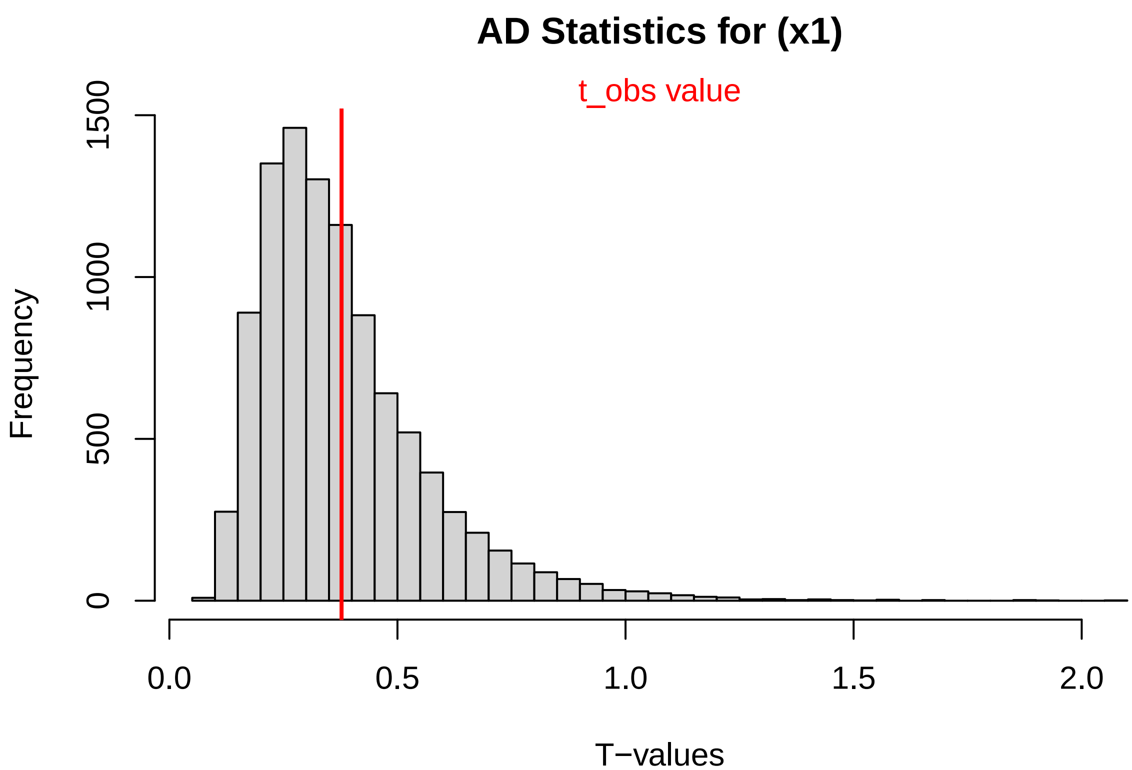
```
## [1] "The p-value for x1 is 0.4062"
```

```r
print(paste("The p-value for x2 is", x2_p_val_results$p_value))
```

```
## [1] "The p-value for x2 is 0.0031"
```

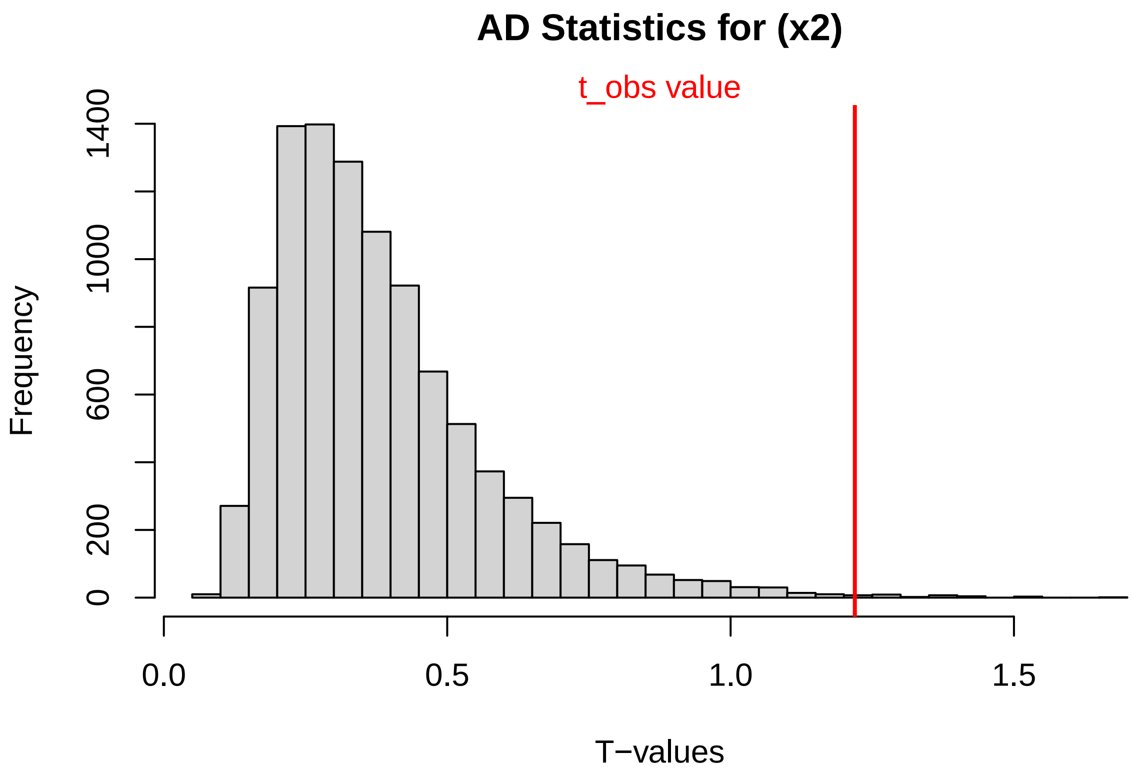
This data is visually represented via a histogram. For x1:

```r
#x1
hist(x1_p_val_results$T_obs_LIST,breaks = 30, main = "AD Statistics for (x1)", xlab = "T-values")
abline(v=x1_p_val_results$t_obs_val, lwd=2, col = "red")
mtext("t_obs value", 3, col='red')
```



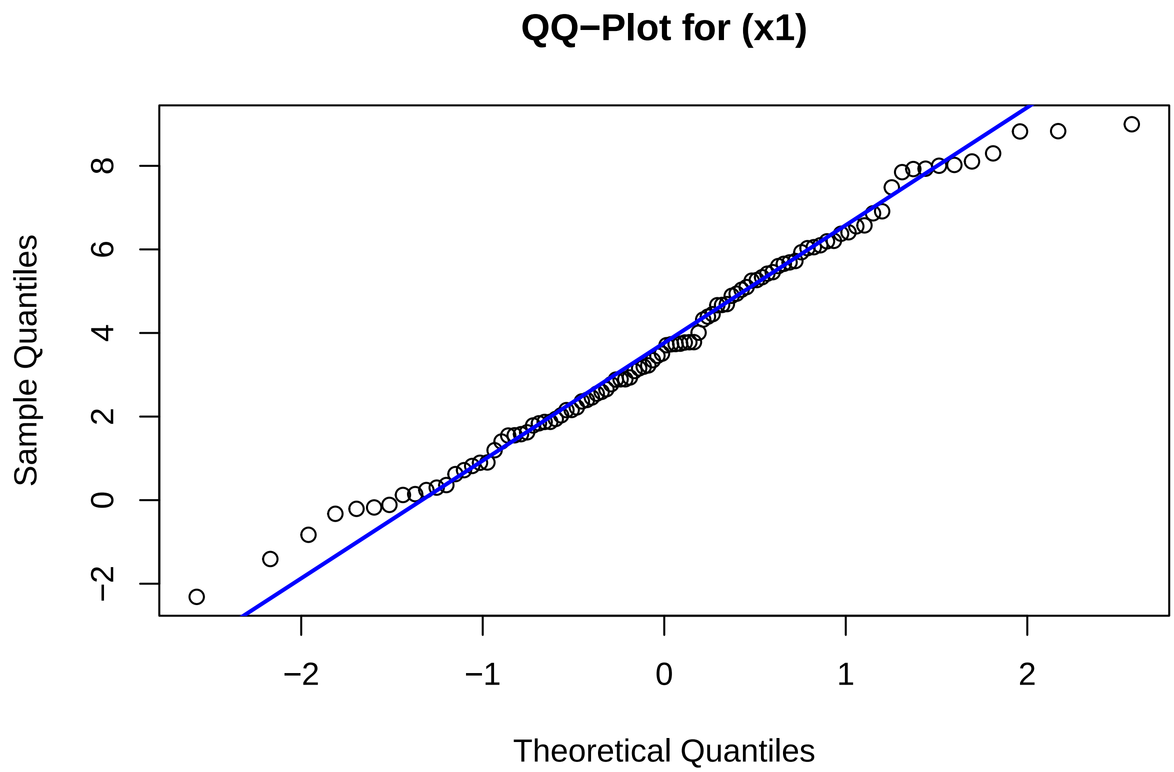
For x2:

```r
#x2
hist(x2_p_val_results$T_obs_LIST, breaks = 30, main = "AD Statistics for (x2)", xlab = "T-values")
abline(v=x2_p_val_results$t_obs_val, lwd = 2, col='red')
mtext("t_obs value", 3,col='red')
```



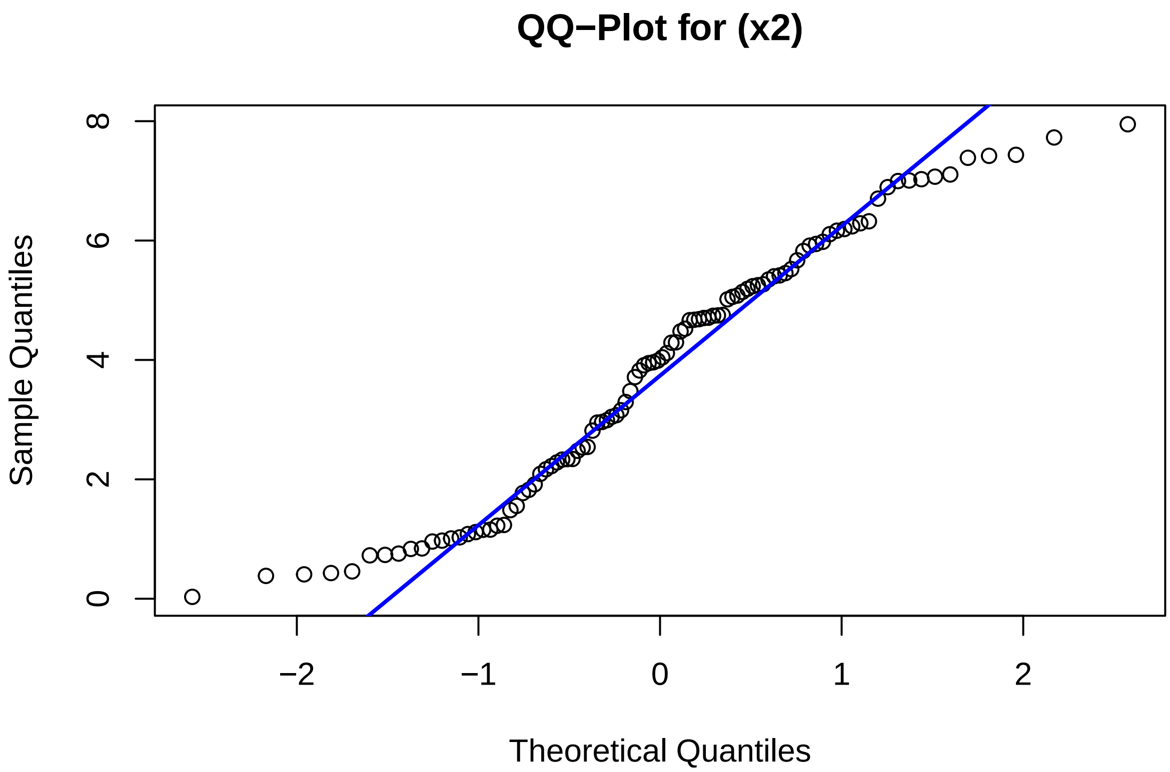
Now, the QQ-plots are considered. For x1:

```
#x1
qqnorm(samples.Q1$x1, main="QQ-Plot for (x1)")
qqline(samples.Q1$x1, col='blue', lwd=2)
```



For x2:

```
#x2
qqnorm(samples.Q1$x2, main = "QQ-Plot for (x2)")
qqline(samples.Q1$x2, col='blue', lwd=2)
```



The samples x1 and x2 can now be analyzed according to the histograms and QQ-plots. In the histogram for data set x1, the t_{obs} value lies within the majority of the simulated distribution. Hence, the deviation from normality is not very large compared to a normal sample of the same size. The QQ-plot reaffirms this analysis because the points plotted are roughly on the line denoting normality. There is no extreme tail behavior or curvature to the graph. Further, the p-value is determined to be ≈ 0.4084. Overall, the p-value is large, the observed statistic is typical for a normal distribution, and the QQ-plot has no major non-linear features. Therefore, **there is no evidence against normality for data set x1**.

Now, consider sample x2. The p-value is found to be ≈ 0.0041. The histogram shows that the observed test statistic does not lie within the range of the bulk of the simulated distribution at all. It seems to lie on the very edge instead. This suggests that there is a deviation from a normal distribution. The QQ-plot reinforces this analysis through the nonlinear points plotted and extreme tail behavior. There is great deviation, especially at the tails, from the normality line plotted, which indicates a non-normal distribution. Overall, the p-value is small, the observed statistic lies on the outskirts of the distribution (which is not expected under the null hypothesis), and the QQ-plot has heavily skewed tails and nonlinear features. Therefore, **there is statistical evidence to reject the null hypothesis that data set x2 is normally distributed.**

e)

The code found above is easily adaptable to different distributions. If a sample were instead compared to an exponential distribution, the following changes would be made:

- The parameter λ for the exponential distribution would be estimated via $\lambda = \frac{1}{\bar{x}}$, where $\bar{x}$ is the mean of the data.
- The sample simulated under the null hypothesis would be generated from an exponential distribution instead of a normal distribution, using the *rexp* command.

Beyond these minor changes, the algorithm remains the same since the Anderson-Darling statistic and p-value are calculated the same way. Therefore, the approach in this question is very general already. It can be used to test a sample against any distribution (provided you can randomly generate samples from the given distribution).

2. Standard Normal Distribution

a)

We are given that $U_1, U_2 \sim U[0,1]$, $\theta \sim U[0, 2*\pi]$, and $R^2 \sim Exp(\frac{1}{2})$. We begin by writing θ in terms of U_2. Since both θ and U_2 follow uniform distributions, scale U_2 appropriately to get:

$$\theta = 2\pi U_2$$

.

To find R in terms of U_1, we must find the inverse for the exponential distribution (since $R^2 \sim Exp(0.5)$). The cumulative distribution function (cdf) for a random variable X in an exponential distribution is: $F_X(x) = 1 - e^{-\lambda x}$. Hence the cdf for R^2 is:

$$F_{R^2}(r^2) = 1 - e^{-0.5r^2}$$

Set $U_1 = F_{R^2}(r^2)$ to apply the inverse transform because U_1 is uniformly distributed.

$$\begin{aligned}
U_1 &= 1 - e^{-0.5r^2} \\
U_1 - 1 &= -e^{-0.5r^2} \\
1 - U_1 &= e^{-0.5r^2} \\
\ln(1 - U_1) &= -\frac{1}{2}r^2 \\
-2\ln(1 - U_1) &= r^2 \\
r &= \sqrt{-2\ln(1 - U_1)}
\end{aligned}$$

Hence, $R = \sqrt{-2\ln(1 - U_1)}$ as required.

b)

Substituting the equations found in part (a) yields expressions for X_1 and X_2 in terms of U_1 and U_2. We find that:

$$X_1 = \sqrt{-2\ln(1 - U_1)}\cos(2\pi U_2)$$

and

$$X_2 = \sqrt{-2\ln(1 - U_1)}\sin(2\pi U_2)$$

as required.

c)

Use properties of polar coordinates to write expressions for U_1 and U_2 in terms of X_1 and X_2.

Use $X_1^2 + X_2^2 = R^2$ identity.

$$\begin{aligned}
X_1^2 + X_2^2 &= (-2\ln(1 - U_1))(\cos^2(2\pi U_2) + \sin^2(2\pi U_2)) \\
&= -2\ln(1 - U_1) \\
\frac{-1}{2}(X_1^2 + X_2^2) &= \ln(1 - U_1) \\
1 - U_1 &= e^{\frac{X_1^2 + X_2^2}{-2}} \\
U_1 &= 1 - e^{\frac{X_1^2 + X_2^2}{-2}}
\end{aligned}$$

Use $\tan(\theta) = \tan(2\pi U_2) = \frac{X_2}{X_1}$ identity.

$$\begin{aligned}
\tan(2\pi U_2) &= \frac{X_2}{X_1} \\
2\pi U_2 &= \arctan(\frac{X_2}{X_1}) \\
U_2 &= \frac{1}{2\pi}\arctan(\frac{X_2}{X_1})
\end{aligned}$$

Hence, $U_1 = 1 - e^{\frac{-1}{2}(X_1^2+X_2^2)}$ and $U_2 = \frac{1}{2\pi}\arctan(\frac{X_2}{X_1})$ as required.

d)

To show the given result, we must set up the problem as it is described. U_1 and U_2 are independent uniform distributions, so they have joint probability density function $f(u_1, u_2) = 1$ where $0 < u_1 < 1$ and $0 < u_2 < 1$.

From part (c),

$$\begin{aligned}
U_1 &= 1 - e^{\frac{1}{-2}(X_1^2+X_2^2)} \\
U_2 &= \frac{1}{2\pi}\arctan(\frac{X_2}{X_1})
\end{aligned}$$

.

Let $X_1 = x_1(U_1, U_2)$ and $X_2 = x_2(U_1, U_2)$, with the single-valued inverses given by $U_1 = u_1(X_1, X_2)$ and $U_2 = u_2(X_1, X_2)$ respectively. Now, we must calculate the Jacobian matrix. Calculate the partial derivatives first.

$$\begin{aligned}
\frac{\partial u_1}{\partial X_1} &= X_1 e^{\frac{-1}{2}(X_1^2+X_2^2)} \\
\frac{\partial u_1}{\partial X_2} &= X_2 e^{\frac{-1}{2}(X_1^2+X_2^2)} \\
\frac{\partial u_2}{\partial X_1} &= \frac{-1}{2\pi}\frac{X_2}{X_1^2+X_2^2} \\
\frac{\partial u_2}{\partial X_2} &= \frac{1}{2\pi}\frac{X_1}{X_1^2+X_2^2}
\end{aligned}$$

Hence, after simplification:

$$\begin{aligned}
\det\begin{pmatrix} \frac{\partial x_1}{\partial U_1} & \frac{\partial x_1}{\partial U_2} \\ \frac{\partial x_2}{\partial U_1} & \frac{\partial x_2}{\partial U_2} \end{pmatrix} &= \frac{\partial x_1}{\partial U_1} \times \frac{\partial x_2}{\partial U_2} - \frac{\partial x_1}{\partial U_2} \times \frac{\partial x_2}{\partial U_1} \\
&= \frac{1}{2\pi(X_1^2+X_2^2)} e^{\frac{-1}{2}(X_1^2+X_2^2)}(X_1^2+X_2^2) \\
&= \frac{1}{2\pi} e^{\frac{-1}{2}(X_1^2+X_2^2)}
\end{aligned}$$

Hence $|J| = \frac{1}{2\pi} exp(\frac{-1}{2}(X_1^2+X_2^2))$. Now, use the joint probability density function of X_1 and X_2 as given by $g(x_1, x_2) = f(u_1, u_2)|J|$.

Then, by the earlier observation about U_1 and U_2, $g(x_1, x_2) = f(u_1, u_2)|J| = (1)|J|$. Therefore,

$$f_{X_1,X_2}(x_1, x_2) = g(x_1, x_2) = \frac{1}{2\pi} e^{\frac{-1}{2}(X_1^2+X_2^2)}$$

as required.

e)

Consider the joint probability density function found in part (d).

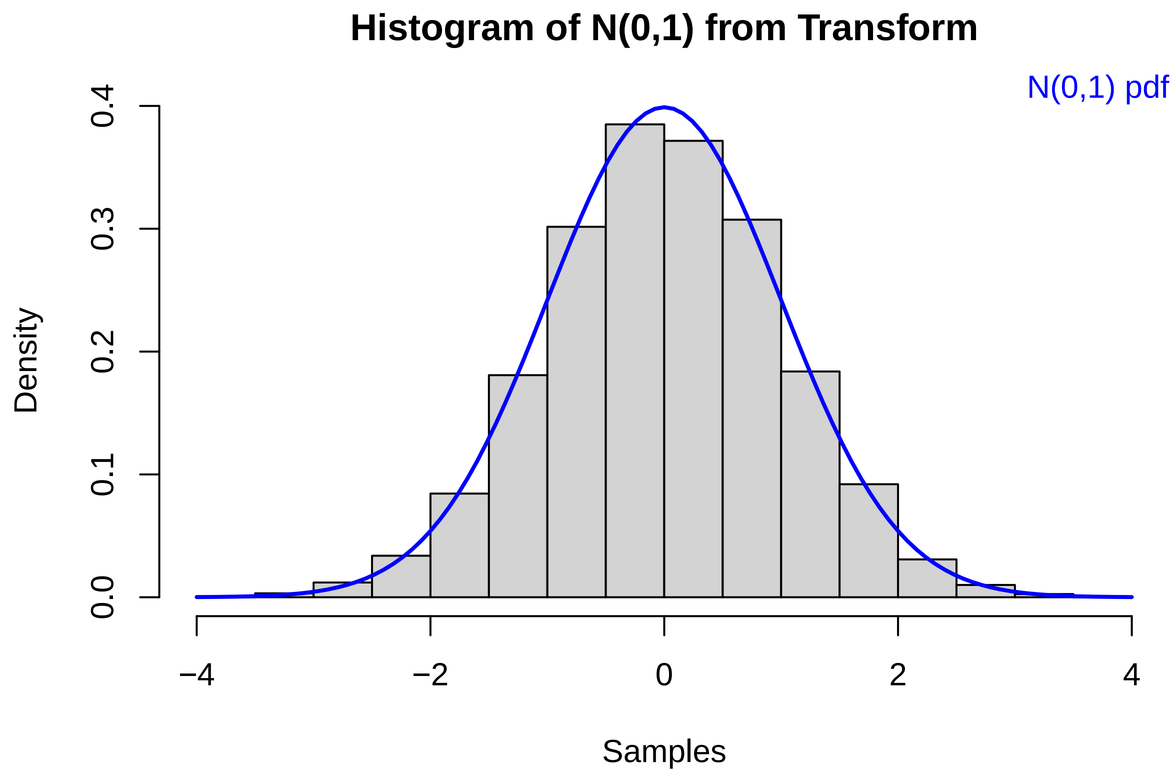
$$\begin{aligned} f_{X_1,X_2}(x_1,x_2) &= \frac{1}{2\pi}e^{\frac{-1}{2}(X_1^2+X_2^2)} \\ &= (\frac{1}{\sqrt{2\pi}}e^{\frac{-1}{2}X_1^2})(\frac{1}{\sqrt{2\pi}}e^{\frac{-1}{2}X_2^2}) \\ &= f_{X_1}(x_1) \times f_{X_2}(x_2) \end{aligned} \tag{1}$$

This is exactly the condition required for independence. Hence, X_1 and X_2 are independent random variables.

f)

To generate 10,000 samples from N(0,1), we take 5000 samples from the X_1 distribution and 5000 from the X_2 distribution. The samples within each distribution are independent by definition and the samples between the two distributions are independent by part (e). The histogram is as follows:

```
#we use both x1 and x2 so together they
#give 10000 samples
num_samples<-5000

U1<-runif(num_samples)
U2 <- runif(num_samples)

#the transform
X1<-sqrt(-2*log(1-U1))*cos(2*pi*U2)
X2<-sqrt(-2*log(1-U1))*sin(2*pi*U2)

X_all<-c(X1,X2)

#histogram
hist(X_all, probability=T, main="Histogram of N(0,1) from Transform", xlab="Samples")
curve(dnorm(x, mean=0,sd=1),lwd=2, add=T,col='blue')
mtext("N(0,1) pdf", 3, adj=1,col='blue')
```



g)

First, we find the Anderson Darling statistic for the above samples:

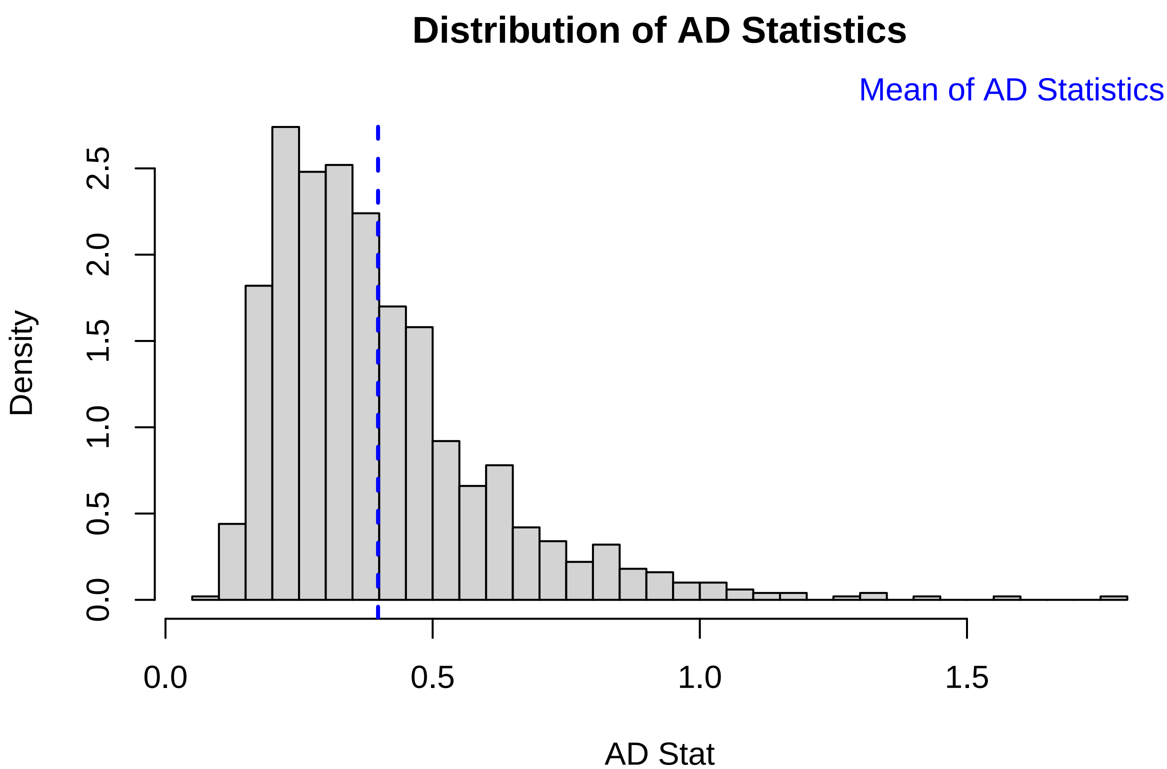
```
ad_stat_here=compute.ad.test(X_all)
print(paste("The Anderson Darling statistic for the above sample is",ad_stat_here))
```

```
## [1] "The Anderson Darling statistic for the above sample is 0.417225492779835"
```

However, to confirm the distribution of the data, these statistics must be found for many samples. We repeat the simulation 1000 times with 100 samples each time. This is seen as follows:

```
#THE ALGORITHM
number_samples<-100
reps<-1000
ad_stats_found<-numeric(reps)

for(i in 1:reps){
  #generate N(0,1)
  U1_rep<-runif(number_samples/2)
  U2_rep<-runif(number_samples/2)

  #transform
  X1_rep<-sqrt(-2*log(1-U1_rep))*cos(2*pi*U2_rep)
  X2_rep<-sqrt(-2*log(1-U1_rep))*sin(2*pi*U2_rep)
```

```
  #sampleset
  set_samples<-c(X1_rep,X2_rep)

  #compute AD stat for each
  ad_stats_found[i]<-compute.ad.test(set_samples)
}

#THE PLOT
hist(ad_stats_found,prob=T, breaks=30, main="Distribution of AD Statistics", xlab="AD Stat")
abline(v=mean(ad_stats_found),col='blue',lwd=2,lty=2)
mtext("Mean of AD Statistics", 3,adj=1,col='blue')
```



From this repeatability test, it is clear that the distribution of the Anderson-Darling Statistics is skewed to the right. The Anderson-Darling Statistic measures how far the sample deviates from a normal distribution. The distribution's right skewedness is an indication that the underlying data is distributed approximately normally. The histogram shows that small deviations from the normal distribution are common, while larger deviations are rarer. Therefore, the distribution of the X-values is approximately normal, as expected.

h)

To generate samples from a general $N(\mu, \sigma^2)$ distribution, a transformation must be applied. We already generated samples $X \sim N(0,1)$. Then, apply an affine transformation such that the new samples, call them Z, satisfy $Z \sim N(\mu, \sigma^2)$. The transformation is given by $Z = \sigma X + \mu$.

Using the univariate version of the earlier result, it is established that if X has pdf $f(x)$ and the transformation $Z = u(X)$ has single-valued inverse $X = v(Z)$, then the pdf of Z is given by:

$$g(z) = f(v(Z)) \times |v'(z)|$$

We have that:

$$Z = \mu + \sigma X$$
$$X = \frac{Z - \mu}{\sigma}$$
$$\frac{dX}{dz} = \frac{1}{\sigma}$$

Hence, $g(z) = f(\frac{Z-\mu}{\sigma})|\frac{1}{\sigma}|$
Because X is normally distributed, we found previously that:

$$f(x) = \frac{1}{\sqrt{(2\pi)}}e^{\frac{-x^2}{2}}$$

Substituting $v(z)$ yields:

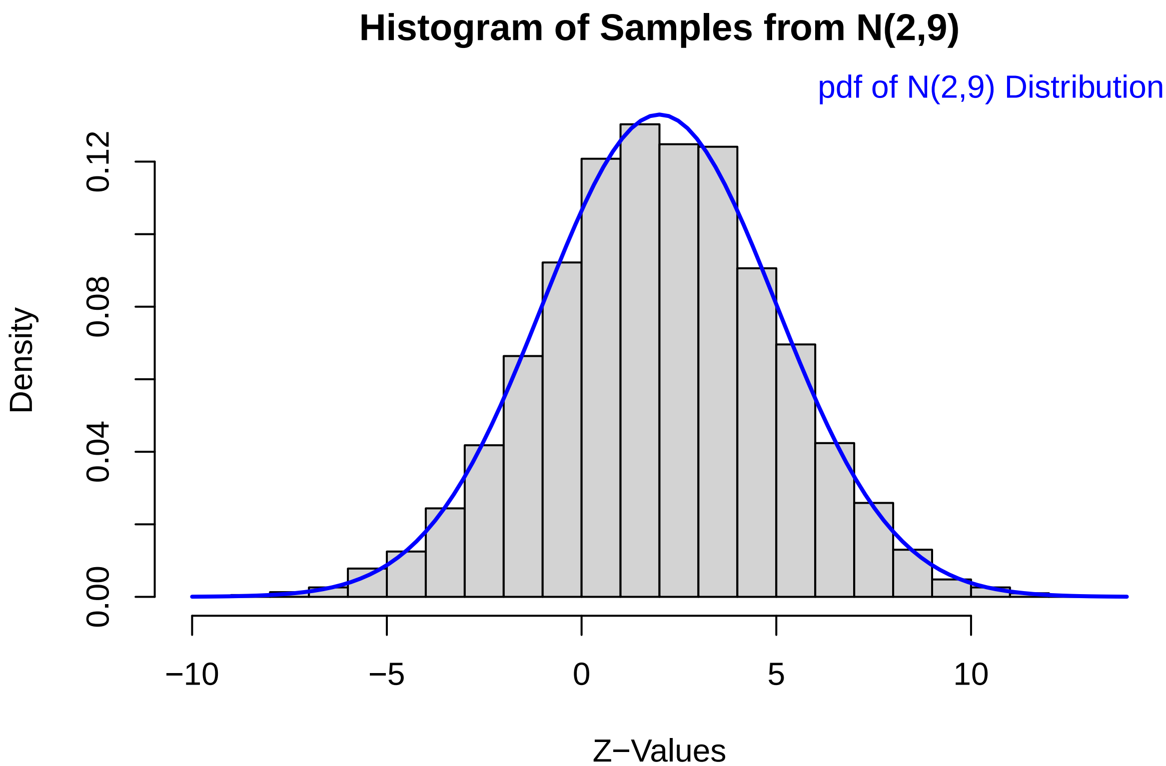
$$g(z) = \frac{1}{\sigma\sqrt{2\pi}}e^{\frac{-(z-\mu)^2}{2\sigma^2}}$$

This is the probability density function (pdf) of the samples generated from a generic $N(\mu, \sigma^2)$ distribution. Computationally, this transformation has a simple implementation because each sample can be adjusted using the original affine transformation $Z = \sigma X + \mu$, given particular values for σ and μ.

i)

From part (f), we have a sample of 10,000 realizations from $N(0, 1)$. Now, we will transform them to be from the $N(2, 9)$ distribution. This is coded as follows:

```
#transform the values
Z_mu <-2
Z_sigma<-3

Z_all<- Z_mu+ Z_sigma*X_all

#plot it
hist(Z_all, breaks=30, probability=T, main="Histogram of Samples from N(2,9)",xlab="Z-Values")
curve(dnorm(x,mean=Z_mu, sd=Z_sigma), col='blue', lwd=2,add=T)
mtext("pdf of N(2,9) Distribution", 3, adj=1, col='blue')
```

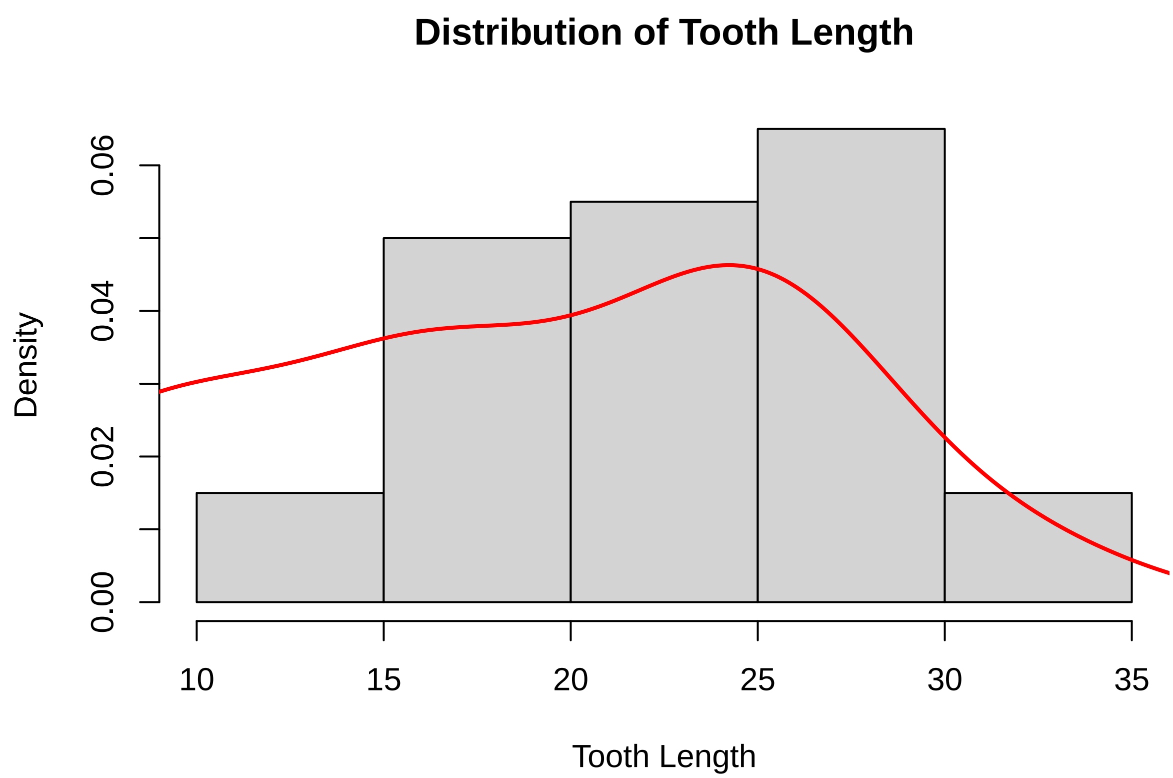


The pdf of the distribution $N(2,9)$ is overlayed to demonstrate the transformation was done appropriately.

3. Exploratory Analysis and Hypothesis Testing

a)i)

```
load("Tooth.Growth.rda")
hist(Tooth.Growth$len, probability=T,xlab = "Tooth Length", main = "Distribution of Tooth Length")
lines(density(ToothGrowth$len), col = "red", lwd = 2)
```

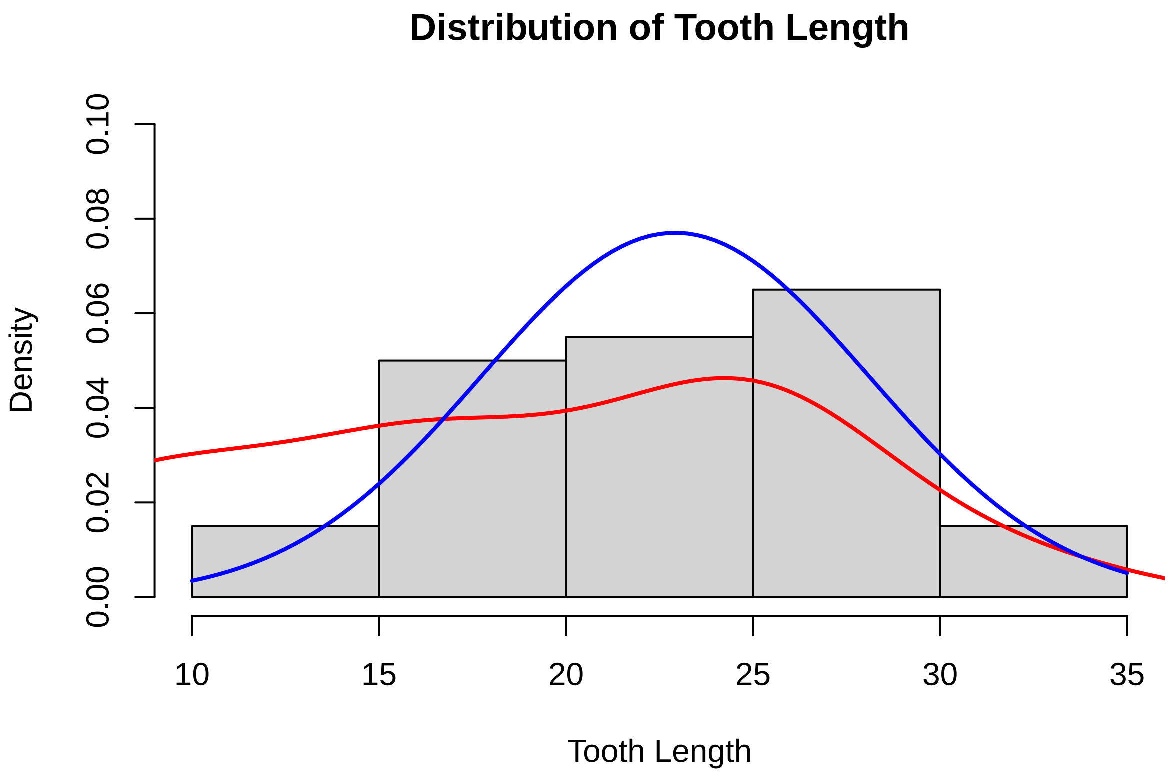


A histogram is the appropriate graphical representation of the Tooth Length because "len" is a continuous numeric variable. Histograms allow for the visualization of the spread and distribution of the data easily. Adding the density of the data also presents more relevant distribution data.

a)ii)

For reference, we superimpose a normal distribution (scaled to appropriate mean and standard deviation values) density curve, shown in blue:

```
load("Tooth.Growth.rda")
#Tooth.Growth
hist(Tooth.Growth$len, probability=T,xlab = "Tooth Length", main = "Distribution of Tooth Length",ylim=c
lines(density(ToothGrowth$len), col = "red", lwd = 2)
curve(dnorm(x, mean = mean(Tooth.Growth$len),sd = sd(Tooth.Growth$len)),col = 'blue',lwd=2, add = T)
```



The distribution appears to be somewhat skewed to the right because there is a higher density on the left than on the right. There are slightly larger tooth lengths found in the right tail than the left. The peak of the curve appears to have shifted right compared to the normal distribution.

The distribution has a moderate peak with slightly lighter tails than a normal distribution. It is hypothesized to be platykurtic. To check, calculate the value of kurtosis:

```
load("Tooth.Growth.rda")
library(moments)
kurtosis(Tooth.Growth$len)
```

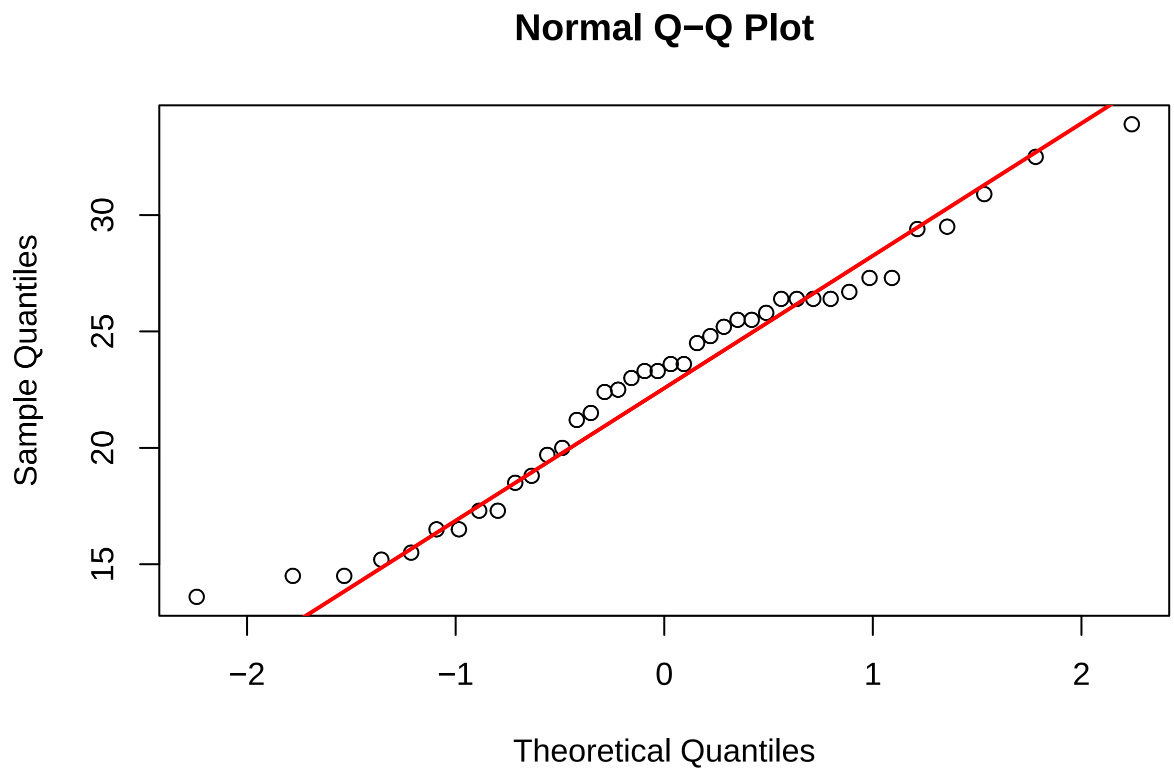
```
## [1] 2.295611
```

This value is slightly less than three, which indicates the distribution is indeed platykurtic (by definition).

a)iii)

From the histogram, the distribution appears to be unimodal and approximately symmetric. There is no apparent extreme skewedness. To further determine whether the distribution of len can be considered normal, generate a QQ-plot.

```
load("Tooth.Growth.rda")
qqnorm(Tooth.Growth$len)
qqline(Tooth.Growth$len, col ="red", lwd =2)
```



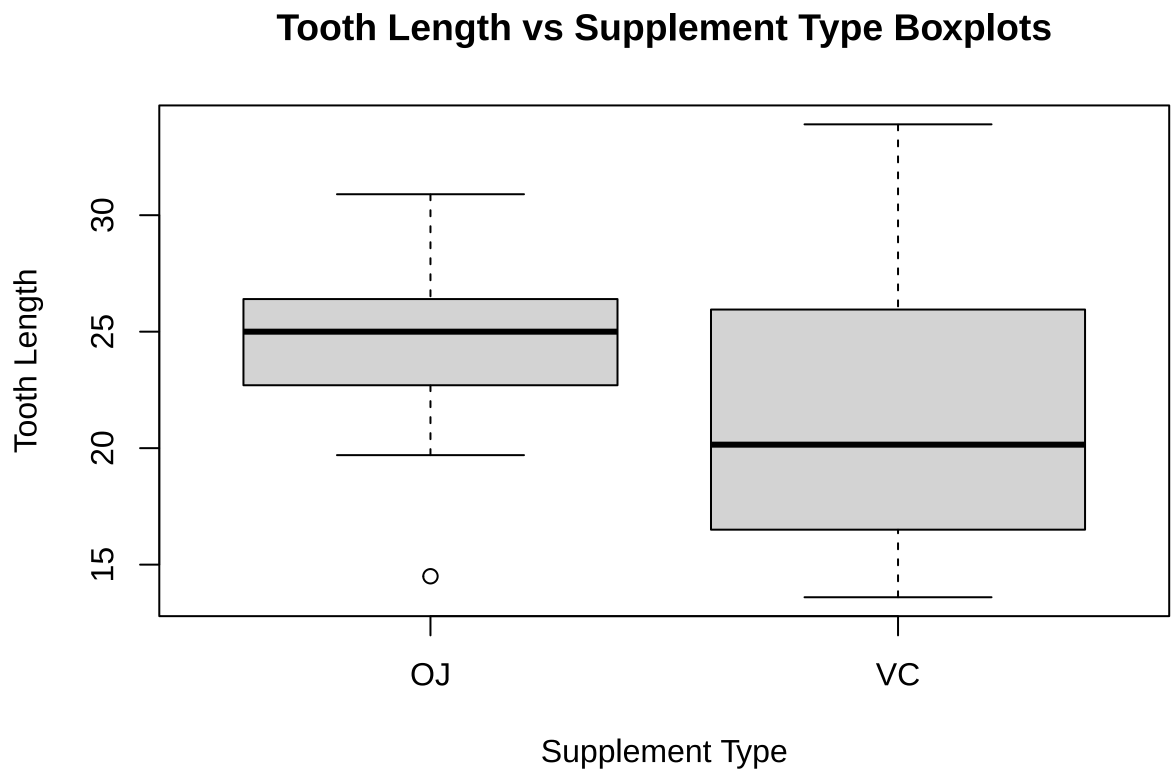
The plot above compares the quantiles from the “len” data to those of a normal distribution. There is a strong linear correlation between these values, as demonstrated by the numerous points that lie near or on the red line calculated. Hence, the Quantile-Quantile Plot and earlier observations suggest that the “len” data can be considered approximately normally distributed.

b) i)

```r
load("Tooth.Growth.rda")
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)

boxplot(Tooth.Growth$len ~ Tooth.Growth$supp,xlab = "Supplement Type", ylab ="Tooth Length" , main= "To
```



The tooth length has a wider range of values for the ascorbic acid supplement than for orange juice. The mean tooth length was slightly higher for the orange juice treatment. There is slight deviation between the two boxplots, indicating that the tooth length may be minorly affected by the delivery method of Vitamin C.

b) ii)

To determine the statistical significance of the deviation discussed in part (bi), a hypothesis test in the form of a F-test must be performed. The orange juice and ascorbic acid groups are independent groups and "len" is a numeric value that takes an approximately normal distribution, so an F-test is the appropriate method.

The *null hypothesis* is that there is no significant statistical difference between the variances of the orange juice and ascorbic acid groups. The *alternate hypothesis* is that there is statistical evidence to show there is a difference in variance between the two groups. This test is completed via the following command:

```
load("Tooth.Growth.rda")
var.test(Tooth.Growth$len~Tooth.Growth$supp)
```

```
##
##  F test to compare two variances
##
## data:  Tooth.Growth$len by Tooth.Growth$supp
## F = 0.36631, num df = 19, denom df = 19, p-value = 0.0342
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.1449918 0.9254760
## sample estimates:
## ratio of variances
##          0.3663147
```

The value of p is less than 0.05 (the default significance level), so the null hypothesis is rejeted. The variances of the two groups cannot be assumed equal. Hence, we can apply a Welch's t-test.

b) iii)

We are then asked to manually compute the above t-test at a 5% significance level.

Model assumptions:

- The guinea pig observations are independent of one another.
- The tooth length ("len") is approximately normally distributed in each supplement group.
- The variances for orange juice and ascorbic acid groups are not assumed to be equal.

Hypotheses:

The *null hypothesis* in this case is that there is no significant statistical difference between the orange juice and ascorbic acid methods on the tooth length (ie: $\mu_{OJ} = \mu_{VC}$ where μ is the mean tooth length for each group). The *alternate hypothesis* is that there is statistical evidence to show supplement type affects tooth growth (ie: $\mu_{OJ} \neq \mu_{VC}$). The alternate hypothesis is chosen to be two-tailed because the test seeks to investigate any impact on tooth length between supplement types. It does not anticipate a particular result, so it cannot be single-tailed.

Test statistic:

The test statistic for this data is:

$$T = \frac{\bar{X}_{OJ} - \bar{X}_{VC}}{\sqrt{\frac{\hat{\sigma}^2_{OJ}}{n_{OJ}} + \frac{\hat{\sigma}^2_{VC}}{n_{VC}}}}$$

. ($\bar{X}_{OJ}$ and $\bar{X}_{VC}$ are the means of the orange juice and ascorbic acid groups, respectively. n_{OJ} and n_{VC} are the number of samples within the orange juice and ascorbic acid groups, respectively. $\hat{\sigma}_{OJ}$ and $\hat{\sigma}_{VC}$ are the variances of the orange juice and ascorbic acid groups, respectively.)

Under H_0, the test statistic should be distributed as $T \sim t_\nu$ with $\nu = 31.273$.

The *observed value* of the test statistic is calculated as follows:

```
load("Tooth.Growth.rda")
OJ<-Tooth.Growth$len[Tooth.Growth$supp=="OJ"]
VC <- Tooth.Growth$len[Tooth.Growth$supp == "VC"]

mu_oj <- mean(OJ)
mu_vc<- mean(VC)
sd_oj <- sd(OJ)
sd_vc<- sd(VC)
n_oj <- length(OJ)
n_vc<- length(VC)

t_obs_sqrt_frac<- ((sd_oj^2)/n_oj)+((sd_vc)^2/n_vc)
t_obs <- (mu_oj-mu_vc)/sqrt(t_obs_sqrt_frac)
print(paste("The t observed value is", t_obs))
```

```
## [1] "The t observed value is 1.83965758981669"
```

The *p-value* can then be calculated:

```
p_val = 2*pt(abs(t_obs), df=31.273, lower.tail=F)
print(paste("The p-value is", p_val))
```

```
## [1] "The p-value is 0.0753283863270034"
```

Then, the critical region for $\alpha = 0.05$, two-tailed test is:

$$t_{crit} = \pm t_{1-\alpha/2,df} = \pm t_{1-0.025,31.273}$$

It is calculated as:

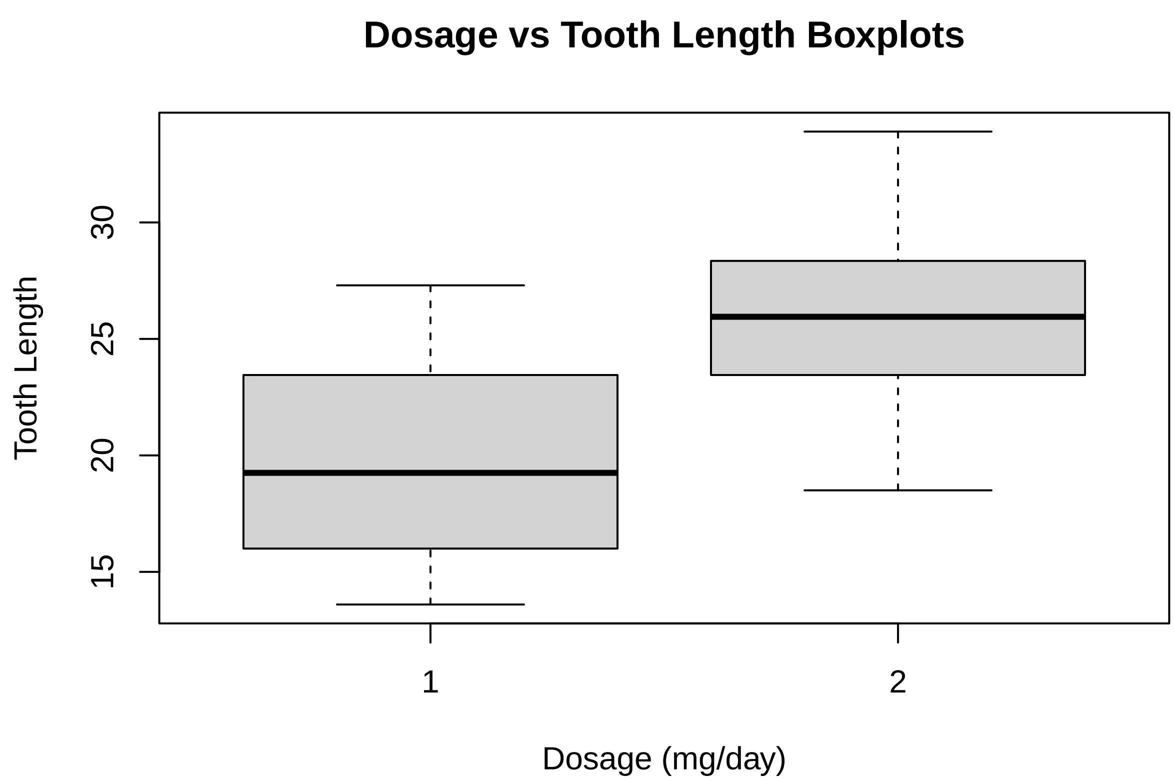
```
t_critical <-qt(1-0.025, 31.273)
t_critical_region = c(-t_critical, t_critical)
print(paste("The critical region is", t_critical_region[1],",", t_critical_region[2]))
```

```
## [1] "The critical region is -2.03879207457099 , 2.03879207457099"
```

Finally, the conclusion drawn from this t-test yields a failure to reject the null hypothesis because $p = 0.075328 > 0.05$ and $|t_{obs}| = 1.839658 < t_{critical} \approx 2.04$. Through both methods of performing a t-test, it is clear that the null hypothesis cannot be rejected. Hence, there is not significant statistical evidence to conclude that the tooth length differs depending on the Vitamin C supplement type.

c) i)

```
load("Tooth.Growth.rda")
boxplot(Tooth.Growth$len ~ Tooth.Growth$dose,xlab = "Dosage (mg/day)", ylab ="Tooth Length", main = "Dos
```



The mean tooth length for the 2 mg/day dose is outside the interquartile range for the 1 mg/day dose. There is very little overlap in ranges between the dosage amounts. From these two boxplots, the tooth length appears to increase substantially as the amount of Vitamin C provided increases.

c) ii)

Similar to the data seen in part (bi), we can utilize an F-test to find the statistical correlation between dosage amount and tooth length. The dosage amount groups are independent and the "len" values are approximately normal, so an independent two-sample t-test is the appropriate method.

Once again, the *null hypothesis* is that there is no significant statistical between variances in the dosage groups. The *alternate hypothesis* is that there is statistical evidence to show the variances are different. The following test is performed using pre-existing R commands.

```
load("Tooth.Growth.rda")
var.test(Tooth.Growth$len~Tooth.Growth$dose, conf.level=0.95)
```

```
##
##  F test to compare two variances
##
## data:  Tooth.Growth$len by Tooth.Growth$dose
## F = 1.3687, num df = 19, denom df = 19, p-value = 0.5005
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.5417489 3.4579584
## sample estimates:
## ratio of variances
##           1.368702
```

The p-value for this test is greater than 0.05 (the default confidence level), so the null hypothesis is accepted. There is not statistical evidence that the amount of Vitamin C within a dose affects the length of a guinea pig's tooth.

c) iii)

Now we are asked to perform the above t-test with a 5% significance level. By the above, a pooled t-test can be used.

Model assumptions:

- The guinea pig observations are independent of one another.
- The tooth length ("len") is approximately normally distributed.
- The variances for 1 mg/day and 2 mg/day groups are assumed to be equal.

Hypotheses:

The *null hypothesis* is that there is no significant statistical difference between the 1 mg/day and 2 mg/day doses on tooth length (ie: $\mu_1 = \mu_2$, where μ represents the mean tooth length for the 1 mg/day group and 2 mg/day group, respectively). The *alternate hypothesis* is there is statistical evidence to show a higher dosage amount affects the tooth growth of the guinea pigs more (ie: $\mu_1 < \mu_2$). The alternate hypothesis is chosen to be one-tailed because the test seeks to investigate a greater impact on tooth length between dosage amounts.

Test statistic:

The test statistic for this data is similar to that of the data in part (biii). It is:

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sigma_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where σ_p is the pooled estimate of the common variance. Since the group sizes are the same, we use the following formula: $\sigma_p^2 = \frac{(n_1-1)\sigma_1^2+(n_2-1)\sigma_2^2}{n_1+n_2-2}$ where σ_1 and σ_2 denote the variances of the 1 mg/day and 2 mg/day dosage groups, respectively. (Further, $\bar{X}_1$ and $\bar{X}_2$ are the mean value of the 1 mg/day and 2 mg/day dosage groups, respectively. n_1 and n_2 are the number of samples in the 1 mg/day and 2 mg/day dosage groups respectively.)

Under the null hypothesis, the test statistic should be distributed as $T \sim t_{n_1+n_2-2}$. The degrees of freedom are: $df = n_1 + n_2 - 2 = 20 + 20 - 2 = 38$. Hence, $T \sim t_{38}$.

The *observed value* of the test statistic is calculated as follows:

```
load("Tooth.Growth.rda")

G1 <-Tooth.Growth$len[Tooth.Growth$dose==1]
G2 <- Tooth.Growth$len[Tooth.Growth$dose==2]

mu_G1 <-mean(G1)
mu_G2 <-mean(G2)
sd_G1 <- sd(G1)
sd_G2<-sd(G2)
n_G1 <- length(G1)
n_G2 <- length(G2)

sigma_p_dose<-sqrt(((n_G1-1)*sd_G1^2+(n_G2-1)*sd_G2^2)/(n_G1+n_G2-2))

Test.stat.frac<- ((1/n_G1) + (1/n_G2))

Test.stat.obs <- (mu_G1-mu_G2)/(sigma_p_dose*sqrt(Test.stat.frac))
print(paste("The observed t-value is", Test.stat.obs))
```

```
## [1] "The observed t-value is -4.90048431719355"
```

The *p-value* can then be calculated:

```
p_value_dosages<-pt(abs(Test.stat.obs), df=38, lower.tail=F)
print(paste("The p-value is", p_value_dosages))
```

```
## [1] "The p-value is 9.05414268090857e-06"
```

Then, the critical region for $\alpha = 0.05$, and a two-tailed test is:

$$t_crit = \pm t_{1-\alpha/2,df} = \pm t_{1-0.025,38} = \pm t_{0.975,38}$$

It is calculated as:

```
T_critical_dosage<-qt(1-0.05, 38)
T_critical_dosage_region <-c(-T_critical_dosage, T_critical_dosage)
print(paste("The critical region is:", T_critical_dosage_region[1], ",", T_critical_dosage_region[2]))
```

```
## [1] "The critical region is: -1.68595446016674 , 1.68595446016674"
```

Finally, the conclusion drawn from this t-test yields a rejection of the null hypothesis because $p \approx 9.05 \times 10^{-6} << 0.05$ and $|t_{obs}| \approx 4.9004 > t_{critical} \approx 1.686$. Through both methods of performing a t-test, it is clear that the null hypothesis can be rejected. Hence, there is significant statistical evidence to conclude that tooth length increases as an effect of a greater amount of Vitamin C supplemented to the guinea pigs.